Unit 3

Types of Random Process

WSS & SSS

A random process $\{X(t)\}$ is wide sense stationary if the following conditions are satisfied

1) $E[X(t)]$ = a constant
2) $E[X(t) \cdot X(t+\tau)]$ a func of τ $(t_1 - t_2)$

Q. A RP $X(t)$ is defined by $\cos \Omega t$ where Ω is a uniform RV in $(0, \omega_0)$. Check whether the process is stationary.

$$f_{\Omega}(\omega) = \frac{1}{\omega_0} \quad 0 \le \omega \le \omega_0$$

$$E[X(t)] = E[\cos \Omega t] = \frac{1}{\omega_0} \int_0^{\omega_0} \cos \Omega t \cdot d\Omega$$

$$= \frac{1}{\omega_0} \left[\frac{\sin \Omega t}{t}\right]_0^{\omega_0} = \frac{1}{\omega_0} \left[\frac{\sin \omega_0 t}{t}\right]$$

$$= \frac{\sin(\omega_0 t)}{\omega_0 t}$$

∴ Not a constant but func of t

∴ $X(t)$ is not stationary

Q. A RP is defined by $x(t) = \cos(t+\phi)$ whr ϕ is a RV with PDF $f(\phi) = \frac{1}{\pi}$, $\frac{-\pi}{2} < \phi < \frac{\pi}{2}$

$$E[x(t)] = E[\cos(t+\phi)]$$

$$= \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t+\phi)\, d\phi$$

$$= \frac{1}{\pi}\left[\sin(t+\phi)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \frac{1}{\pi}\left[\sin\left(t+\frac{\pi}{2}\right) - \sin\left(t-\frac{\pi}{2}\right)\right]$$

$\cos t + \cos t$

$= \frac{2\cos t}{\pi}$ → Not const but func of t

∴ Not stationary

Q. A RP is given by $x(t) = A\sin(\omega_0 t)$ whr ω_0 is a const & A is a RV uniformly distributed over -1 to 1. PT its not stationary

$f_A(A) = 1/2 \quad -1 < A < 1$

① $E[x(t)] = E[A \cdot \sin\omega_0 t] = \sin(\omega_0 t) \cdot E[A]$

$$= \frac{1}{2}\int_{-1}^{1} A \cdot dA = 0 \to \text{const}$$

$$R_x(\tau) = E[x(t) \cdot x(t+\tau)]$$

$$= E[A\sin(\omega_0 t) \cdot A\sin(\omega_0(t+\tau))]$$

$$= \sin(\omega_0 t) \cdot \sin(\omega_0(t+\tau)) \cdot E[A^2]$$

$$= \quad \frac{1}{2}\int_{-1}^{1} A^2\, dA$$

= not a func of τ → $x(t)$ is not stationary classmate

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Q. ST RP given by $X(t) = A\cos(\omega_0 t + \theta)$ whr A, ω_0 are consts & θ is uniformly distri over 0 to 2π is a WSS

PDF $\rightarrow f_\theta(\theta) = \frac{1}{2\pi} \quad 0 < \theta < 2\pi$

① $E[X(t)] = E[A\cos(\omega_0 t + \theta)]$

$$= \frac{1}{2\pi} \cdot A \cdot \int_0^{2\pi} \cos(\omega_0 t + \theta)\, d\theta$$

$$= \frac{A}{2\pi}\left[\sin(\omega_0 t + \theta)\right]_0^{2\pi}$$

$$= \frac{A}{2\pi}\left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t)\right]$$

$$= \frac{A}{2\pi}\left[\sin(\omega_0 t) - \sin(\omega_0 t)\right] = 0$$

② $R_X(\tau) = E[X(t) \cdot X(t+\tau)]$

$$= E[A\cos(\omega_0 t + \theta) \cdot A\cos(\omega_0(t+\tau) + \theta)]$$

$$= \frac{A^2}{2\pi}\int_0^{2\pi} \cos(\omega_0 t + \theta) \cdot \cos(\omega_0(t+\tau) + \theta)\, d\theta$$

$$= \frac{A^2}{2}\cos\omega_0\tau$$, a function of τ

$\therefore X(t)$ is WSS

→ Evolutionary Process
- A process which is not stationary
- In other words, A RP is evolutionary if mean & var are function of t. (dep on t)

Ergodicity

If $X(t)$ is a RP, then $\frac{1}{2T}\int_{-T}^{T} x(t)\,dt$ is called time avg

A RP is ergodic if time avg $(\bar{X}_t)$ = Ensemble avg $E[X(t)]$

Q. A RP is given by $X(t) = A\cos(\omega_0 t + \theta)$ where θ is a RV uniformly distri over $[0, 2\pi]$. ST ~~the~~ process is mean-ergodic

- Ensemble avg $\rightarrow$ $E[X(t)] = E[A\cos(\omega_0 t + \theta)]$

$$= \frac{A}{2\pi}\int_0^{2\pi} \cos(\omega_0 t + \theta)\,d\theta = \frac{A}{2\pi}\left[\sin(\omega_0 t + \theta)\right]_0^{2\pi}$$

$$= \frac{A}{2\pi}\left[\sin(2\pi + \omega_0 t) - \sin(\omega_0 t)\right]$$

$$= \frac{A}{2\pi}\left[\sin(\omega_0 t) - \sin(\omega_0 t)\right] = 0$$

- Time avg

$$\bar{X}_T = \frac{1}{2T}\int_{-T}^{T} x(t)\,dt = \frac{A}{2T}\int_{-T}^{T} \cos(\omega_0 t + \theta)\,dt$$

$$= \frac{A}{2T}\left[\frac{\sin(\omega_0 t + \theta)}{\omega_0}\right]_{-T}^{T} = \frac{A}{2T}\left[\frac{\sin(\omega_0 T + \theta)}{\omega_0} - \frac{\sin(\theta - \omega_0 T)}{\omega_0}\right]$$

Since $\sin(\omega_0 T \pm \theta)$ is bounded as $T \rightarrow \infty$

$$\lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t)\, dt = 0$$

$\therefore$ Time avg = Ensemble avg = 0

$\therefore$ $x(t)$ is mean ergodic

Q. A RP given by $X(t) = \cos(t+\phi)$ where ϕ is a RV $U \sim [0, 2\pi]$.

(i) ST process is stationary in WS

(ii) ST process is mean ergodic

- Ensemble avg $\rightarrow E[X(t)]$

(i) $$E[X(t)] = E[\cos(t+\phi)]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \cos(t+\phi)\, d\phi$$

$$= \frac{1}{2\pi} \left[\cos(t+\phi)\right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[\cos(t+2\pi) - \cos(t)\right]$$

$$= 0 \rightarrow \text{const}$$

$$R_x(\tau) = E[X(t) \cdot X(t+\tau)]$$

$$= E\left[\cos(t+\phi) \cdot \cos(t+\phi+\tau)\right]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \cos(t+\phi) \cdot \cos(t+\phi+\tau)\, d\phi$$

$$= \frac{1}{4\pi} \int_0^{2\pi} \cos(2t+2\phi+\tau) + \cos(\tau)\, d\phi$$

$$= \frac{1}{4\pi}\left[\frac{\sin(2t+2\phi+\tau)}{2} + \phi\cos(\tau)\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left[\frac{\sin(4\pi+(2t+\tau))}{2} + 2\pi\cos(\tau) - \frac{\sin(2t+\tau)}{2}\right]$$

$$= \frac{1}{\cancel{4\pi}\,2}\left[\frac{\sin(2t+\tau)}{\cancel{2}} - \frac{\sin(2t+\tau)}{\cancel{2}} + \cancel{2\pi}\cos(\tau)\right]$$

$$= \frac{\cos(\tau)}{2} \rightarrow \text{func of } \tau$$

$\therefore$ $x(t)$ is wss

(ii) $$\frac{1}{2T}\int_{-T}^{T}\cos(t+\phi)\,dt$$

$$= \frac{1}{2T}\left[\sin(t+\phi)\right]_{-T}^{T} \rightarrow \frac{1}{2T}\left[\sin(T+\phi)+\sin(T-\phi)\right]$$

$$= \frac{1}{\cancel{2}T}\left[\cancel{2}\cos(T)\cdot\sin(\phi)\right]$$

$$= \frac{1}{T}\left[\cos(T)\cdot\sin(\phi)\right]$$

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}x(t)\,dt = 0$$

$\therefore$ Time avg = Ensemble avg = 0

$\therefore$ Mean ergodic

Q. A RP is given by $X(t) = A\cos\omega t + B\sin\omega t$, where A & B are RV such that $E[A] = 0$, $E[B] = 0$, $E[AB] = 0$, $E[A^2] = E[B^2]$. PT process is mean ergodic

→ Markov Chain

A RP in which the set of states Space S & set of parameter T are discrete taking only a finite or countable no. of finite values in which the probablity P_{ij}, that the process will go from i^{th} state to j^{th} state in the next step depends only on the present state & not on how it arrived at that state is called Markov Chain of the first order.

$$P[X_{n+1} = j \mid X_0 = i_0, X_1 = i_1, X_2 = i_2, \ldots X_{n-1} = i_{n-1}, \ldots X_n = i]$$
$$= P[X_{n+1} = j \mid X_n = i]$$

$X_0, X_1, \ldots X_{n-1}$ → past value

X_n → present value

X_{n+1} → future value

One step Transition Probablity Matrix

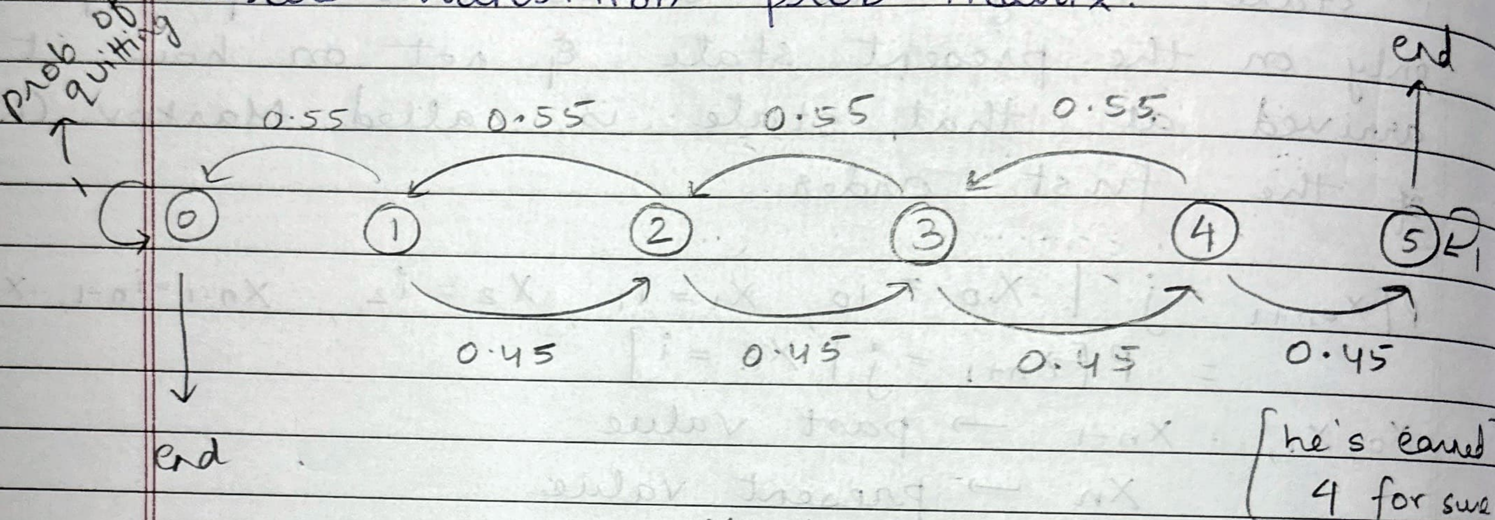
States of X_{n+1}

$$\begin{array}{ccc} & & \begin{array}{cccc} 0 & 1 & 2 & \cdots \end{array} \\ \text{States of } X_n & \begin{array}{c} 0 \\ 1 \\ 2 \\ \vdots \end{array} & \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ P_{20} & P_{21} & P_{22} & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix} \end{array}$$

Rules

1. $P_{ij} \geq 0$; $i, j \geq 0$
2. $\sum\limits_{j=0} P_{ij} = 1$; $i = 0, 1, \ldots$

Q. Suppose in a gambling game, a gambler wins Rs. 1 with prob 0.45 & loses Rs. 1 with prob 0.55 at any state. He starts the game with Rs. 1 & quits when either he's earned Rs. 4 or he's bankrupt. Write transition prob matrix.

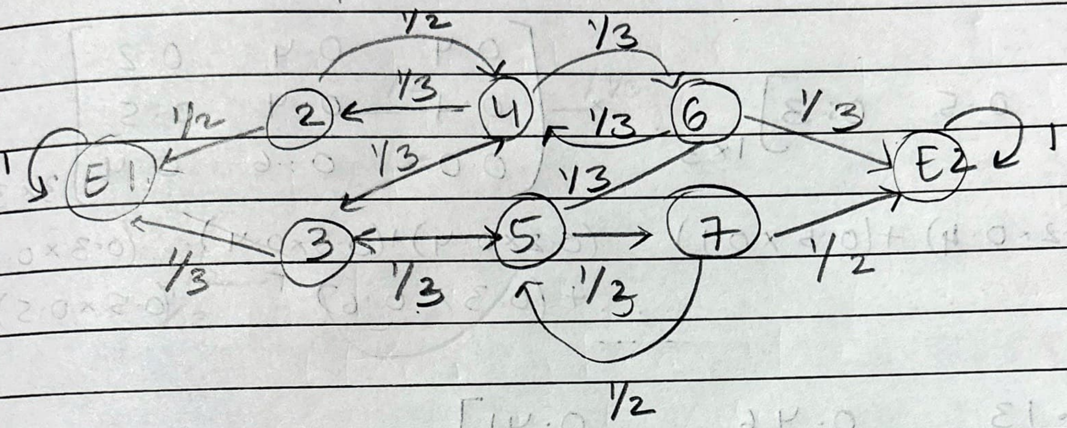


$$X_{n+1}$$

$$X_n \quad \begin{array}{c|cccccc} & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0.55 & 0 & 0.45 & 0 & 0 & 0 \\ 2 & 0 & 0.55 & 0 & 0.45 & 0 & 0 \\ 3 & 0 & 0 & 0.55 & 0 & 0.45 & 0 \\ 4 & 0 & 0 & 0 & 0.55 & 0 & 0.45 \\ 5 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}$$

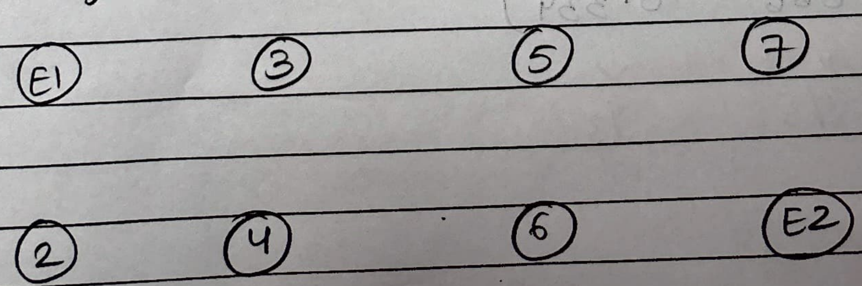
Q. Consider an exp in which a rat is wandering inside a maze which has 8 compartments ~~E1, E2~~ Each compartment has doors as shown in figure. If the rat reaches compartment E1 / E2, it escapes else it moves from one compartment to other through doors at random. Write matrix & figure.

E1 1	3	5	7
2	4	6	E2 1



$$
\begin{array}{c}
 \\ E_1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ E_2
\end{array}
\begin{array}{c}
\begin{array}{cccccccc} E_1 & 2 & 3 & 4 & 5 & 6 & 7 & E_2 \end{array} \\
\begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 \\
1/3 & 0 & 0 & 1/3 & 1/3 & 0 & 0 & 0 \\
0 & 1/3 & 1/3 & 0 & 0 & 1/3 & 0 & 0 \\
0 & 0 & 1/3 & 0 & 0 & 1/3 & 1/3 & 0 \\
0 & 0 & 0 & 1/3 & 1/3 & 0 & 0 & 1/3 \\
0 & 0 & 0 & 0 & 1/2 & 0 & 0 & 1/2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
\end{array}
$$

Better Diagram



Q. $\begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.1 & 0.4 & 0.5 \\ 0.0 & 0.6 & 0.4 \end{bmatrix}$ $\mu(1) = 0.2$
$\mu(2) = 0.5$
$\mu(3) = 0.3$

Find probs of the process at $T = 1$

$$\begin{bmatrix} 0.2 & 0.5 & 0.3 \end{bmatrix}_{1\times 3} \times \begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.1 & 0.4 & 0.5 \\ 0.0 & 0.6 & 0.4 \end{bmatrix}_{3\times 3}$$

$= (0.2\times 0.4) + (0.5\times 0.1) \quad (0.2\times 0.4) + (0.5\times 0.4) + (0.3\times 0.6) \quad (0.3\times 0.2) + (0.5\times 0.5) + (0.3\times 0$

$= \begin{bmatrix} 0.13 & 0.46 & 0.41 \end{bmatrix}$

→ Steady State Probablities (Limiting Prob)

TPM

$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$

$\pi = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \end{bmatrix}$

$P^2 = \begin{bmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{bmatrix}$

$P_i(n) \to \pi_i$ as $n \to \infty$

$P^3 = \begin{bmatrix} 0.667 & 0.333 \\ 0.666 & 0.334 \end{bmatrix}$

$\Sigma \pi_i P = \pi_i$ & $\Sigma \pi_i = 1$

↓ $2/3$ ↓ $1/3$

Q. If TPM of Markov Chain is:

$$P = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix}$$ Find its steady state matrix → upto power 8

$$P^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix}, \quad P^3 = \begin{bmatrix} 0.25 & 0.75 \\ 0.375 & 0.625 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} 0.375 & 0.625 \\ 0.3125 & 0.6875 \end{bmatrix}, \quad P^5 = \begin{bmatrix} 0.3125 & 0.6875 \\ 0.3437 & 0.6562 \end{bmatrix}$$

$$P^6 \dots, \quad P^7 \dots, \quad P^8 = \begin{bmatrix} 0.3359 & 0.664 \\ 0.332 & 0.6679 \end{bmatrix}$$

$$\pi = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

→ Poisson Process

$$P[X(t) = x] = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots$$

Mean $= \lambda t =$ Var

ACR → $R_X(t_1, t_2) = E[X(t_1) \cdot X(t_2)] = \lambda^2 t_1 t_2 + \lambda t_1$

ACV → $C_X(t_1, t_2) = R_X(t_1, t_2) - E[X(t_1)] \cdot E[X(t_2)]$
$= \lambda t_1$

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Q. If a bank receives an avg of 6 bad cheques per day (i) What are the prob that it'll receive 4 bad cheques on any given day. (ii) 10 bad cheques on any 2 consequetive days

(i) $\lambda = 6$

$t = 1 \quad \rightarrow \quad \lambda t = 6$

$x = 4$

$$E[X(1) = 4] = \frac{e^{-6} \cdot (6)^4}{4!} = 0.134$$

(ii) $t = 2 \quad \rightarrow \quad \lambda t = 12$

$x = 10$

$$P(X(2) = 10) = \frac{e^{-12} \cdot (12)^{10}}{10!} = 0.105$$

Q. Customers arrive at a counter in accordance w Poisson process w mean rate of 2 per min. Find prob that the interval b/w 2 consequetive arrivals is (i) > 1 min, (ii) b/w 1 & 2 mins, (iii) ≤ 4 mins

By property of Poisson process, the time of interval t b/w 2 consequetive customers follows expo d

$\lambda = 2, \quad t = 1$

$f(t) = \lambda e^{-\lambda t} = 2e^{-2t}$

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(i) $P(t>1) = \int_1^{\infty} 2e^{-2t}\, dt \rightarrow 2\left[\frac{e^{-2t}}{-2}\right]_1^{\infty}$

$= -1\left[0 - e^{-2}\right] = \boxed{0.135}$

(ii) $P(1<t<2) = 2\int_1^{2} e^{-2t}\, dt \longrightarrow -1\left[e^{-2t}\right]_1^{2}$

$= -e^{-4} + e^{-2} = \boxed{0.117}$

(iii) $P(t \leq 4) = 2\int_0^{4} e^{-2t}\, dt \longrightarrow -1\left[e^{-2t}\right]_0^{4}$

$= -e^{-8} + e^{1} \rightarrow 0.999$

Q. A radioactive substance emits particles at rate of 6 particles per min accd to poisson process. The prob that emitted particles will be rec is 0.5. Find the prob that 10 particles will be recorded in 4 mins time period.

$$P[N(t)=r] = \frac{e^{-\lambda p t} \cdot (\lambda p t)^r}{r!}$$

$\lambda = 6,\ p = 0.5,\ x = 10,\ t = 4$

$$= \frac{e^{-6 \cdot 0.5 \cdot 4} \cdot (6 \cdot 0.5 \cdot 4)^{10}}{10!}$$

$= \boxed{0.104}$